

BHAVYA KANSAL

AI/ML Engineer · Deep Learning Researcher · Computer Vision & Gen AI

Patiala, Punjab, India | +91 6283332944 | kansalbhavya27@gmail.com | bhavyakansal.dev | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Deep Learning Researcher, currently in 2nd year of B.Tech (AI & Data Science) at Thapar Institute of Engineering and Technology (TIET), having completed a Diploma in Computer Science Engineering at Thapar Polytechnic College (June 2026) and a 6-month AI/ML internship at NIELIT Ropar in collaboration with IIT Ropar. Demonstrates strong command of end-to-end AI product development — from model design, training, and optimization to full-stack deployment on cloud platforms. Has independently delivered production-grade AI systems spanning healthcare diagnostics, computer-vision restoration, and agentic multimodal assistants — all live and publicly accessible. Open to research collaborations, industry internships, and high-impact AI/ML roles.

EDUCATION

Thapar Institute of Engineering and Technology (TIET)

B.Tech in Data Science and Artificial Intelligence

2026 – Present

Patiala, Punjab

Thapar Polytechnic College

Diploma in Computer Science Engineering

August 2023 – June 2026

Patiala, Punjab

EXPERIENCE

IIT Ropar × NIELIT Ropar — Joint Collaboration Program

January 2026 – July 2026

AI/ML Internship

Ropar, Punjab

- Completed a 6-month AI/ML internship at NIELIT Ropar in active collaboration with IIT Ropar's premier innovation and research center, building production-ready AI systems in deep learning, computer vision, NLP, and multimodal AI.
- Delivered multiple AI projects using multimodal agentic intelligence, including a multi-disease healthcare AI platform (Healthy AI), as program deliverables; mentored by Dr. Sarwan Singh, Nikshep Paliwal, and Lovnish Verma (NIELIT Ropar).

PROJECTS

Healthy AI — Multi-Disease Medical AI Platform

- Built and deployed a production-ready multi-module medical AI platform spanning heart disease prediction, diabetes risk assessment (ensemble ML with clinical feature engineering), and brain tumor MRI classification (4 tumor types, 91.62% accuracy via fine-tuned deep CNN).
- Deployed on Render with a professional dark-themed Flask web interface.

GitHub: github.com/BhavyaKansal20/healthy-ai | Live: <https://healthyai-dlog.onrender.com/>

ChromaCrystal UHD — AI Photo Colorization & Restoration Platform

- Built an AI-powered computer vision web app that colorizes and restores black-and-white photographs via a FastAPI-driven 5-stage pipeline: preprocessing, DDColor colorization, GFPGAN face restoration, Real-ESRGAN upscaling, and cinematic enhancement.
- Developed a Next.js 15 (TypeScript) frontend with a dark neon UI and WebSocket-based real-time job tracking; deployed on Hugging Face Spaces.

GitHub: github.com/BhavyaKansal20/ChromaCrystal_UHD | Live: <https://bhavyakansal20-chromacrystal-uhd.hf.space>

Agni AI — Sovereign Multimodal AI Assistant

- Built “Agni” (Advance Augmented Generative Nano Intelligent AI assistant) — India's sovereign multimodal AI assistant delivering context-aware conversational intelligence across Telegram and web.
- Engineered a Next.js + PostgreSQL/Prisma ORM backend integrating Sarvam AI's multimodal models with Google/GitHub OAuth; built a Tailwind CSS/Framer Motion web app alongside the Telegram bot.

Telegram: t.me/AgniAIBot | Live: <https://aagni-ai.veercel.app>

TECHNICAL SKILLS

Languages: Python (Expert), HTML/CSS/JS, Java (Intermediate)

ML / DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, OpenCV, MediaPipe

Deep Learning: CNN, MobileNetV2, EfficientNet-B0, LSTM, BiLSTM, GAN (Basic), Transformer architecture

NLP & GenAI: NLP pipelines, sequence modeling, LLM fine-tuning, RAG, ReAct agent patterns

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, data preprocessing pipelines

Deployment & Tools: Hugging Face Spaces, Render, Vercel, Docker, Git, VS Code, Colab, Kaggle

RESEARCH & ACHIEVEMENTS

- AI in Healthcare: Researching diabetic retinopathy grading via fundus image deep learning and explainable AI (XAI) for clinical decision support.
- Generative AI & Multimodal Systems: Researching PEFT/LoRA fine-tuning, RAG pipelines, and cross-modal (vision + audio + text) fusion for production deployment.